

A/D and D/A converters



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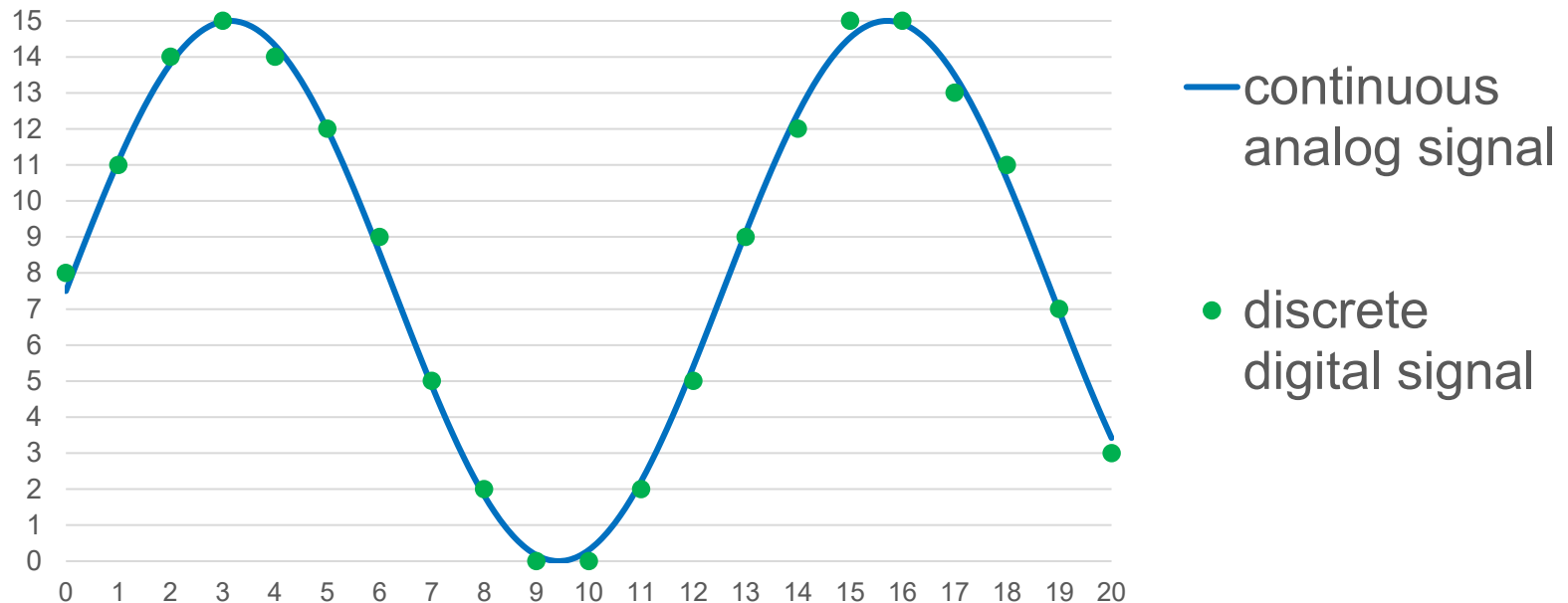


Analog and Digital conversion

- Computers interact with real world
 - analog signals in input (e.g., temperature measured by a sensor)
 - analog signals in output (e.g., playing a song).
- Processed values are always digital.
- An analog-to-digital converter (ADC) receives a signal with an infinite amount of values and convert it into a binary representation.
- A digital-to-analog converter (DAC) turns a binary code into an analog voltage.

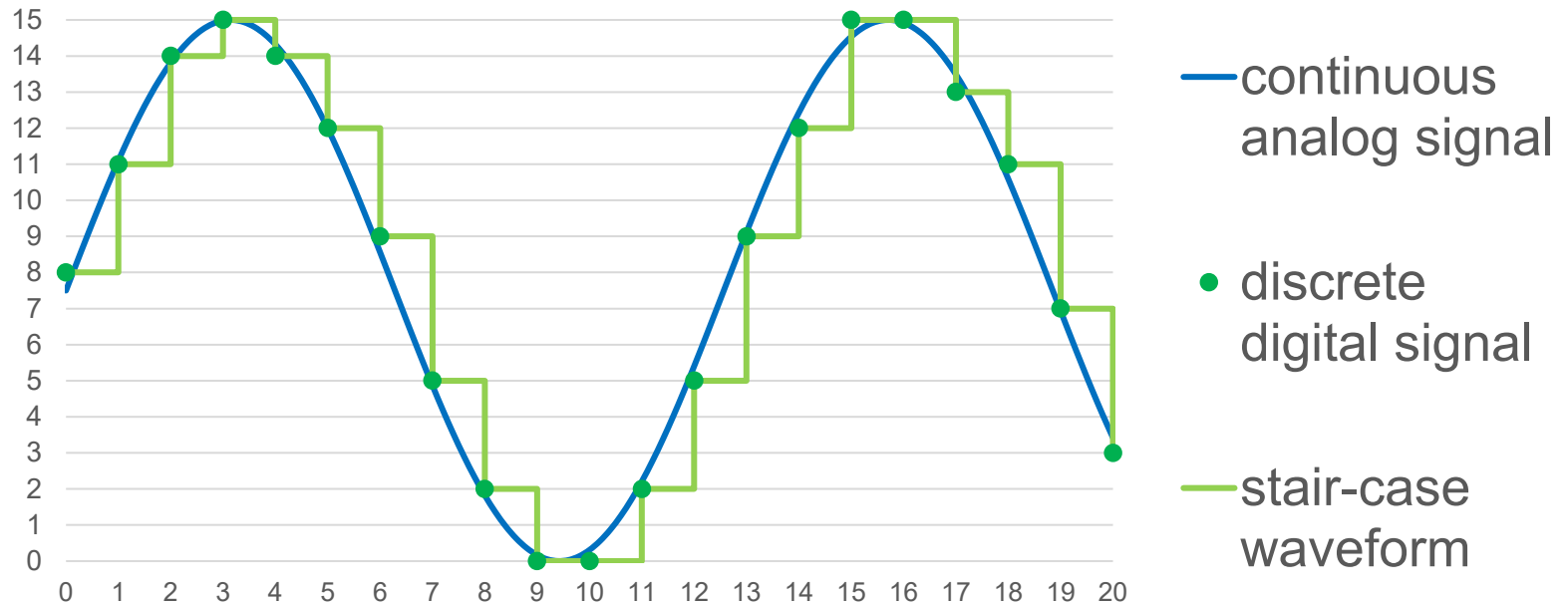
Digitization of analog signals

1. Sampling: few points are chosen on the analog signal.
2. Rounding: digital values are approximately equal to the analog values.



Digitization of analog signals

3. Digital values are joined to round off the analog signal to near stabilized values.
- Such a process is called Time and Amplitude Quantization.

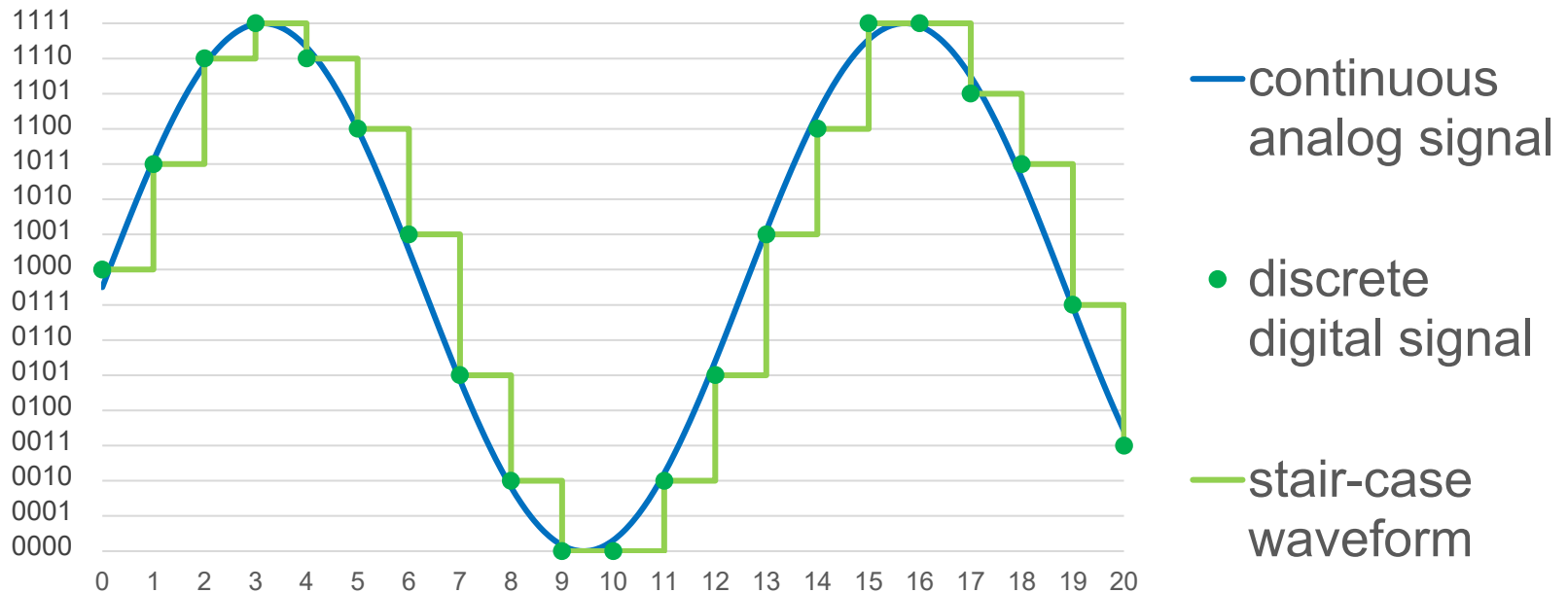


Time quantization

- The analog signal is sampled at discrete points in time.
- The time between two discrete points is referred to as the sampling interval.
- The minimum sample interval is the inverse of the maximum conversion rate of the ADC
 - they depend on the specific ADC circuit.

Amplitude quantization

- The analog signal is discretized with a fixed number of quantization levels.
- The resolution is specified in bits; it indicates the number of distinct output codes (2^n).



A numerical example

- From the datasheet:
 - the resolution is $n = 4$
 - the full-scale input range is $0 - 3 \text{ V}$.
- The ADC can encode an analog input into $2^4 = 16$ different levels
 - 3 V corresponds to 15 (maximum ADC code)
 - lower input voltages are mapped proportionally.
- The quantum or quantization step (spacing between two adjacent representation levels) is $3 \text{ V} / 16 = 0.1875 \text{ V}$

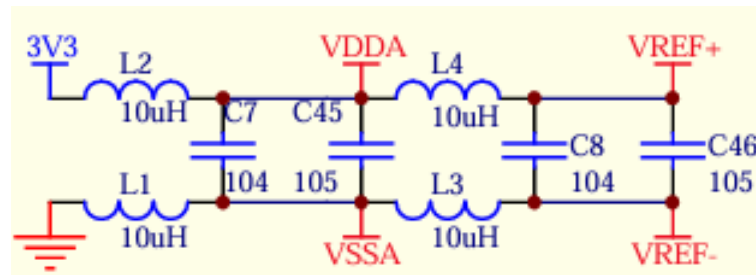
On-chip ADC in LPC1768

- 12-bit successive approximation ADC
- Input multiplexing among 8 pins AD0.0-AD0.7

U1							
CAN1_RX	P0.0	46	P0.0/RD1/TXD3/SDA1	P1.0/ENET_TXD0	95	P1.0	ENET_TXD0
CAN1_TX	P0.1	47	P0.1/TD1/RXD3/SCL1	P1.1/ENET_TXD1	94	P1.1	ENET_TXD1
UART0_TX	P0.2	98	P0.2/TXD0/AD0.7	P1.4/ENET_TX_EN	93	P1.4	ENET_TX_EN
UART0_RX	P0.3	99	P0.3/RXD0/AD0.6	P1.8/ENET_CRS	92	P1.8	ENET_CRS
CAN2_RX	P0.4	81	P0.4/I2SRX_CLK/RD2/CAP2.0	P1.9/ENET_RXD0	91	P1.9	ENET_RXD0
CAN2_TX	P0.5	80	P0.5/I2SRX_WS/TD2/CAP2.1	P1.10/ENET_RXD1	90	P1.10	ENET_RXD1
TP_CS	P0.6	79	P0.6/I2SRX_SDA/SSEL1/MAT2.0	P1.14/ENET_RX_ER	89	P1.14	ENET_RX_ER
SCK1	P0.7	78	P0.7/I2STX_CLK/SCK1/MAT2.1	P1.15/ENET_REF_CLK	88	P1.15	ENET_REF_CLK
MISO1	P0.8	77	P0.8/I2STX_WS/MISO1/MAT2.2	P1.16/ENET_MDC	87	P1.16	ENET_MDC
MOSI1	P0.9	76	P0.9/I2STX_SDA/MOSI1/MAT2.3	P1.17/ENET_MDIO	86	P1.17	ENET_MDIO
UART2_TX	P0.10	48	P0.10/TXD2/SDA2/MAT3.0	P1.18/USB_UP_LED/PWM1.1/CAP1.0	32	P1.18	USB_UP_LED
UART2_RX	P0.11	49	P0.11/RXD2/SCL2/MAT3.1	P1.19/MC0A/USB_PPWR/CAP1.1	33	P1.19	USB_PPWR
SCK	P0.15	62	P0.15/TXD1/SCK0/SCK	P1.20/MCFB0/PWM1.2/SCK0	34	P1.20	SCK0
DF_CS	P0.16	63	P0.16/RXD1/SSEL0/SSEL	P1.21/MCABORT/PWM1.3/SSEL0	35	P1.21	SD_CS
MISO	P0.17	61	P0.17/CTS1/MISO0/MISO	P1.22/MC0B/USB_PWRD/MAT1.0	36	P1.22	USB_PWRD
MOSI	P0.18	60	P0.18/DCD1/MOSI0/MOSI	P1.23/MCFB1/PWM1.4/MISO0	37	P1.23	MISO0
LCD_EN	P0.19	59	P0.19/DSR1/SDA1	P1.24/MCFB2/PWM1.5/MOSI0	38	P1.24	MOSI0
LCD_LE	P0.20	58	P0.20/DTR1/SCL1	P1.25/MC1A/MAT1.1	39	P1.25	JOY_SEL
LCD_DIR	P0.21	57	P0.21/RI1/RD1	P1.26/MC1B/PWM1.6/CAP0.0	40	P1.26	JOY_DOWN
LCD_CS	P0.22	56	P0.22/RTS1/TD1	P1.27/CLKOUT/USB_OVRCR/CAP0.1	43	P1.27	JOY_LEFT
LCD_RS	P0.23	9	P0.23/AD0.0/I2SRX_CLK/CAP3.0	P1.28/MC2A1.0/MAT0.0	44	P1.28	JOY_RIGHT
LCD_WR	P0.24	8	P0.24/AD0.1/I2SRX_WS/CAP3.1	P1.29/MC2B/PCAP1.1/MAT0.1	45	P1.29	JOY_UP
LCD_RD	P0.25	7	P0.25/AD0.2/I2SRX_SDA/TXD3	P1.30/VBUS/AD0.4	21	P1.30	VBUS
AOUT	P0.26	6	P0.26/AD0.3/AOUT/RXD3	P1.31/SCK1/AD0.5	20	P1.31	ADC_IN
SDA	P0.27	25	P0.27/SDA0/USB_SDA				
SCL	P0.28	24	P0.28/SCL0/USB_SCL	P2.0/PWM1.1/TXD1	75	P2.0	DB00
USB_D+	P0.29	29	P0.29/USB_D+	P2.1/PWM1.2/RXD1	74	P2.1	DB01
USB_D-	P0.30	30	P0.30/USB_D-	P2.2/PWM1.3/CTS1/TRACEDATA3	73	P2.2	DB02
					70	P2.3	

Voltage references

- Measurement range from V_{REFN} to V_{REFP}
 - the Negative Voltage Reference V_{REFN} is lower-bounded by the Voltage at the Source V_{SS} (0V)
 - the Positive Voltage Reference V_{REFP} is upper-bounded by the Voltage at the Drain V_{DD} (3V)



A/D registers (page 587)

Table 531. ADC registers

Generic Name	Description	Access	Reset value ⁽¹⁾	AD0 Name & Address
ADCR	A/D Control Register. The ADCR register must be written to select the operating mode before A/D conversion can occur.	R/W	1	AD0CR - 0x4003 4000
ADGDR	A/D Global Data Register. This register contains the ADC's DONE bit and the result of the most recent A/D conversion.	R/W	NA	AD0GDR - 0x4003 4004
ADINTEN	A/D Interrupt Enable Register. This register contains enable bits that allow the DONE flag of each A/D channel to be included or excluded from contributing to the generation of an A/D interrupt.	R/W	0x100	AD0INTEN - 0x4003 400C
ADDR0	A/D Channel 0 Data Register. This register contains the result of the most recent conversion completed on channel 0.	RO	NA	AD0DR0 - 0x4003 4010
ADDR1	A/D Channel 1 Data Register. This register contains the result of the most recent conversion completed on channel 1.	RO	NA	AD0DR1 - 0x4003 4014
ADDR2	A/D Channel 2 Data Register. This register contains the result of the most recent conversion completed on channel 2.	RO	NA	AD0DR2 - 0x4003 4018
ADDR3	A/D Channel 3 Data Register. This register contains the result of the most recent conversion completed on channel 3.	RO	NA	AD0DR3 - 0x4003 401C
ADDR4	A/D Channel 4 Data Register. This register contains the result of the most recent conversion completed on channel 4.	RO	NA	AD0DR4 - 0x4003 4020
ADDR5	A/D Channel 5 Data Register. This register contains the result of the most recent conversion completed on channel 5.	RO	NA	AD0DR5 - 0x4003 4024
ADDR6	A/D Channel 6 Data Register. This register contains the result of the most recent conversion completed on channel 6.	RO	NA	AD0DR6 - 0x4003 4028
ADDR7	A/D Channel 7 Data Register. This register contains the result of the most recent conversion completed on channel 7.	RO	NA	AD0DR7 - 0x4003 402C
ADSTAT	A/D Status Register. This register contains DONE and OVERRUN flags for all of the A/D channels, as well as the A/D interrupt/DMA flag.	RO	0	AD0STAT - 0x4003 4030
ADTRM	ADC trim register.	R/W	0x0000 0F00	AD0TRM - 0x4003 4034

A/D Control Register (page 588)

Table 532: A/D Control Register (AD0CR - address 0x4003 4000) bit description

Bit	Symbol	Value	Description	Reset value
7:0	SEL		Selects which of the AD0.7:0 pins is (are) to be sampled and converted. For AD0, bit 0 selects Pin AD0.0, and bit 7 selects pin AD0.7. In software-controlled mode, only one of these bits should be 1. In hardware scan mode, any value containing 1 to 8 ones is allowed. All zeroes is equivalent to 0x01.	0x01
15:8	CLKDIV		The APB clock (PCLK_ADC0) is divided by (this value plus one) to produce the clock for the A/D converter, which should be less than or equal to 13 MHz. Typically, software should program the smallest value in this field that yields a clock of 13 MHz or slightly less, but in certain cases (such as a high-impedance analog source) a slower clock may be desirable.	0
16	BURST	1	The AD converter does repeated conversions at up to 200 kHz, scanning (if necessary) through the pins selected by bits set to ones in the SEL field. The first conversion after the start corresponds to the least-significant 1 in the SEL field, then higher numbered 1-bits (pins) if applicable. Repeated conversions can be terminated by clearing this bit, but the conversion that's in progress when this bit is cleared will be completed. Remark: START bits must be 000 when BURST = 1 or conversions will not start. If BURST is set to 1, the ADGINTEN bit in the AD0INTEN register (Table 534) must be set to 0.	0
		0	Conversions are software controlled and require 65 clocks.	
20:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
21	PDN	1	The A/D converter is operational.	0
		0	The A/D converter is in power-down mode.	
23:22	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	START		When the BURST bit is 0, these bits control whether and when an A/D conversion is started:	0
		000	No start (this value should be used when clearing PDN to 0).	
		001	Start conversion now.	

Conversion rate

- The APB clock (PCLK_ADC0) is divided by $(CLKDIV + 1)$ to produce the A/D clock
 - CLKDIV corresponds to bits 15:8 in AD0CR
- The A/D clock should be ≤ 13 MHz
- A conversion requires 65 clocks
- The conversion rate is $13 \text{ MHz} / 65 = 200 \text{ kHz}$

Operating modes

- BURST = 0 and START = 001: software-triggered conversion
- BURST = 0 and START contains 010-111: hardware-triggered conversion
 - on a specified edge of 2 input pins: P2.10 or P2.11
 - on a specified edge of 4 Timer Match signal: MAT0.1, MAT0.3, MAT1.0, MAT1.1
- BURST = 1: burst conversion
 - repeated conversions scanning through the pins selected by bits set to ones in the SEL field.

Power-down mode

- On reset, the ADC is disabled. To enable it:
 - first, set the PCADC bit in the PCONP register
 - then, set the PDN bit in the AD0CR register

Bit	Symbol	Description	Reset value
0	-	Reserved.	NA
1	PCTIM0	Timer/Counter 0 power/clock control bit.	1
2	PCTIM1	Timer/Counter 1 power/clock control bit.	1
3	PCUART0	UART0 power/clock control bit.	1
4	PCUART1	UART1 power/clock control bit.	1
5	-	Reserved.	NA
6	PCPWM1	PWM1 power/clock control bit.	1
7	PCI2C0	The I ² C0 interface power/clock control bit.	1
8	PCSPI	The SPI interface power/clock control bit.	1
9	PCRTC	The RTC power/clock control bit.	1
10	PCSSP1	The SSP 1 interface power/clock control bit.	1
11	-	Reserved.	NA
12	PCADC	A/D converter (ADC) power/clock control bit.	0

Note: Clear the PDN bit in the AD0CR before clearing this bit, and set this bit before setting PDN.

A/D Data Registers

- 8 A/D Channel Data Registers: each one holds the result of the last conversion on the corresponding channel.
- 1 A/D Global Data Register: it holds the result of the most recent A/D conversion that has completed (regardless of the channel).
- All registers have also a DONE flag: this bit is set to 1 when a conversion completes; it is cleared when the register is read.

A/D Global Data Register (p. 589)

Bit	Symbol	Description	Reset value
3:0	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:4	RESULT	When DONE is 1, this field contains a binary fraction representing the voltage on the AD0[n] pin selected by the SEL field, as it falls within the range of V_{REFP} to V_{REFN} . Zero in the field indicates that the voltage on the input pin was less than, equal to, or close to that on V_{REFN} , while 0xFFF indicates that the voltage on the input was close to, equal to, or greater than that on V_{REFP} .	NA
23:16	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
26:24	CHN	These bits contain the channel from which the RESULT bits were converted (e.g. 000 identifies channel 0, 001 channel 1...).	NA
29:27	-	Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
30	OVERRUN	This bit is 1 in burst mode if the results of one or more conversions was (were) lost and overwritten before the conversion that produced the result in the RESULT bits. This bit is cleared by reading this register.	0
31	DONE	This bit is set to 1 when an A/D conversion completes. It is cleared when this register is read and when the ADCR is written. If the ADCR is written while a conversion is still in progress, this bit is set and a new conversion is started.	0

- The bits of the 8 Channel Data Registers are the same, except bits 26:24, which are reserved.

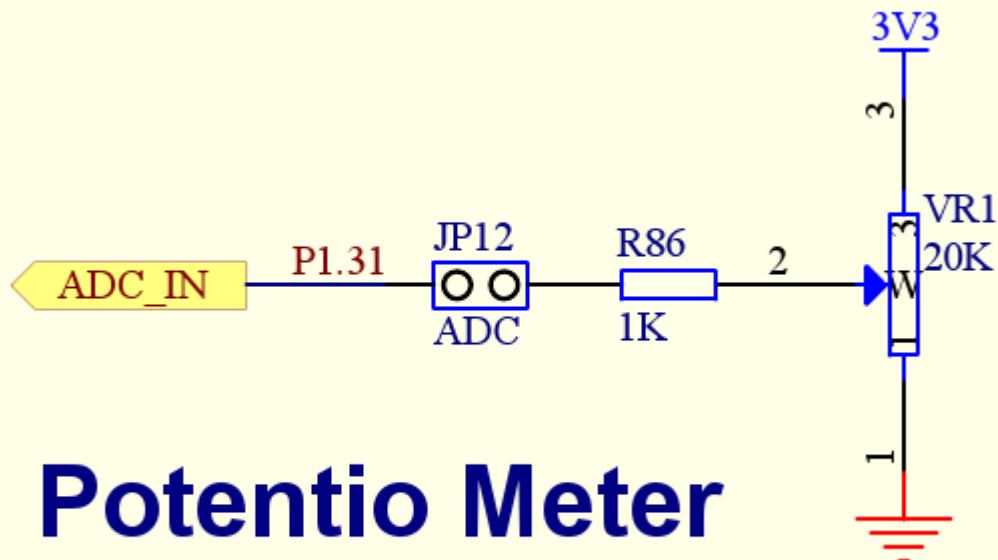
A/D Interrupt Enable register (p. 589)

Table 534: A/D Interrupt Enable register (AD0INTEN - address 0x4003 400C) bit description

Bit	Symbol	Value	Description	Reset value
0	ADINTEN0	0	Completion of a conversion on ADC channel 0 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 0 will generate an interrupt.	
1	ADINTEN1	0	Completion of a conversion on ADC channel 1 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 1 will generate an interrupt.	
2	ADINTEN2	0	Completion of a conversion on ADC channel 2 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 2 will generate an interrupt.	
3	ADINTEN3	0	Completion of a conversion on ADC channel 3 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 3 will generate an interrupt.	
4	ADINTEN4	0	Completion of a conversion on ADC channel 4 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 4 will generate an interrupt.	
5	ADINTEN5	0	Completion of a conversion on ADC channel 5 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 5 will generate an interrupt.	
6	ADINTEN6	0	Completion of a conversion on ADC channel 6 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 6 will generate an interrupt.	
7	ADINTEN7	0	Completion of a conversion on ADC channel 7 will not generate an interrupt.	0
		1	Completion of a conversion on ADC channel 7 will generate an interrupt.	
8	ADGINTEN	0	Only the individual ADC channels enabled by ADINTEN7:0 will generate interrupts. Remark: This bit must be set to 0 in burst mode (BURST = 1 in the AD0CR register).	1
		1	Only the global DONE flag in ADDR is enabled to generate an interrupt.	
31:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

ADC connection on board

- Adjustable potentiometer VR1 is connected to analog channel P1.31.
- JP12 jumper enables the potentiometer input.
- VR1 provides input voltages between 0 V and 3 V to the ADC through the pin AD0.5.



init() and start_conversion()

```
7 void ADC_init (void) {
8
9     LPC_PINCON->PINSEL3 |= (3UL<<30);      /* P1.31 is AD0.5
10
11     LPC_SC->PCONP        |= (1<<12);        /* Enable power to ADC block
12
13     LPC_ADC->ADCR         = (1<< 5) |        /* select AD0.5 pin
14                          (4<< 8) |          /* ADC clock is 25MHz/5
15                          (1<<21);          /* enable ADC
16
17     LPC_ADC->ADINTEN      = (1<< 8);         /* global enable interrupt
18
19     NVIC_EnableIRQ(ADC_IRQn);               /* enable ADC Interrupt
20 }
21
22 void ADC_start_conversion (void) {
23     LPC_ADC->ADCR |= (1<<24);               /* Start A/D Conversion
24 }
```

ADC_IRQHandler()

```
50 void ADC_IRQHandler(void) {  
51  
52     unsigned short AD_current;  
53     unsigned short AD_last = 0xFF;      /* Last converted value  
54  
55     AD_current = ((LPC_ADC->ADGDR>>4) & 0xFFF); /* Read Conversion Result  
56     if(AD_current != AD_last){  
57         /* your action here */  
58         AD_last = AD_current;  
59     }  
60     ADC_start_conversion();  
61 }
```



Sampled value is read

On-chip DAC in LPC1768

- 10-bit Digital to Analog Converter (DAC)
- Precision = 1024 levels between 0 and 3.3V
- Maximum update rate of 1 MHz
- It has no control bit in the PCONP register
 - to enable the DAC, its output must appear on pin P0.26, by configuring the PINSEL1 register.
 - setting the direction of P0.26 seems needed only for software debugging: for functions other than GPIO, the direction is controlled automatically with the board (page 118 of the user manual).

D/A Converter Register (page 594)

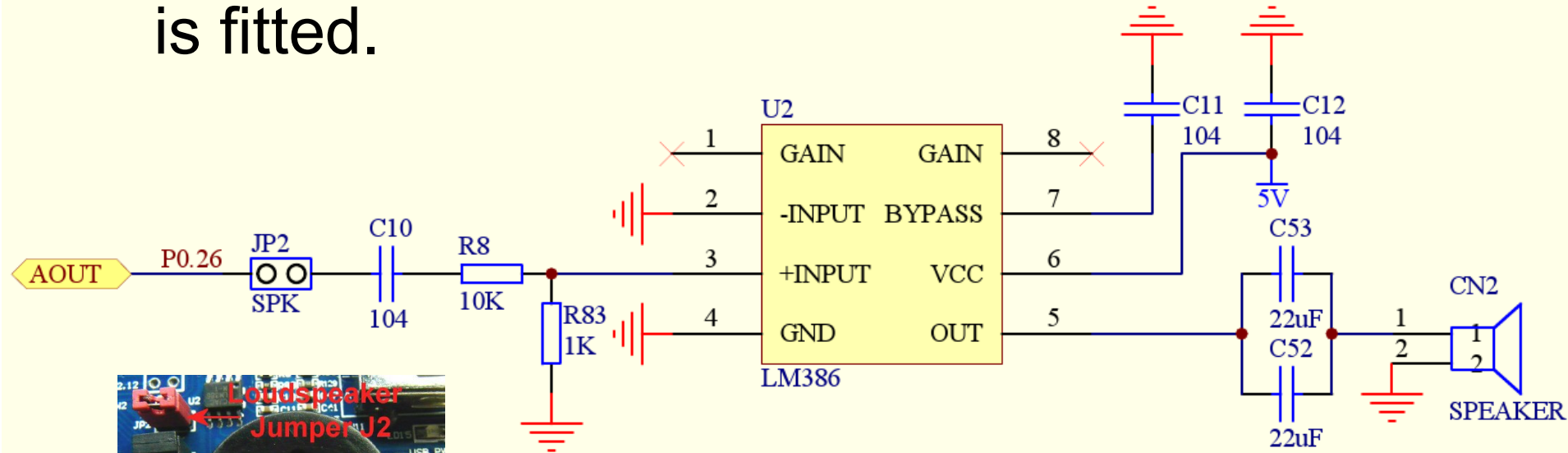
- It contains the 10-bit digital value to be converted to analog and a power control bit.

Table 540: D/A Converter Register (DACR - address 0x4008 C000) bit description

Bit	Symbol	Value	Description	Reset Value
5:0	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA
15:6	VALUE		After the selected settling time after this field is written with a new VALUE, the voltage on the AOUT pin (with respect to V_{SSA}) is $VALUE \times ((V_{REFP} - V_{REFN})/1024) + V_{REFN}$.	0
16	BIAS [1]	0	The settling time of the DAC is 1 μ s max, and the maximum current is 700 μ A. This allows a maximum update rate of 1 MHz.	0
		1	The settling time of the DAC is 2.5 μ s and the maximum current is 350 μ A. This allows a maximum update rate of 400 kHz.	
31:17	-		Reserved, user software should not write ones to reserved bits. The value read from a reserved bit is not defined.	NA

DAC connection on board

- The DAC output is connected to the input of the external speaker (pin P0.26).
- The speaker is enabled when the JP2 jumper is fitted.



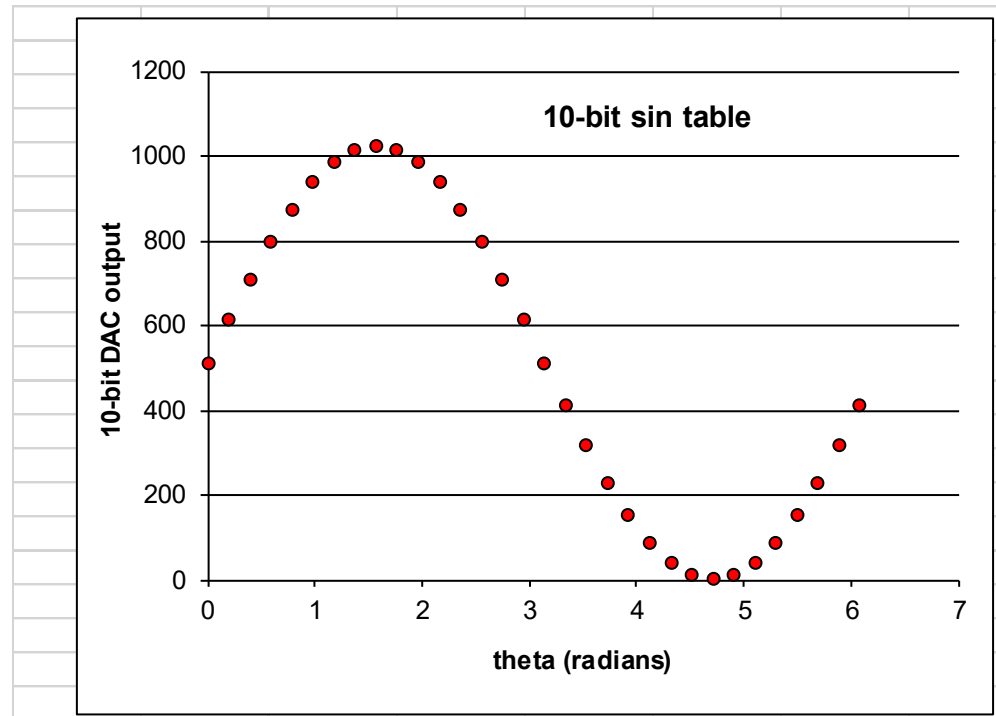
Speaker



10-bit Sinusoid Table

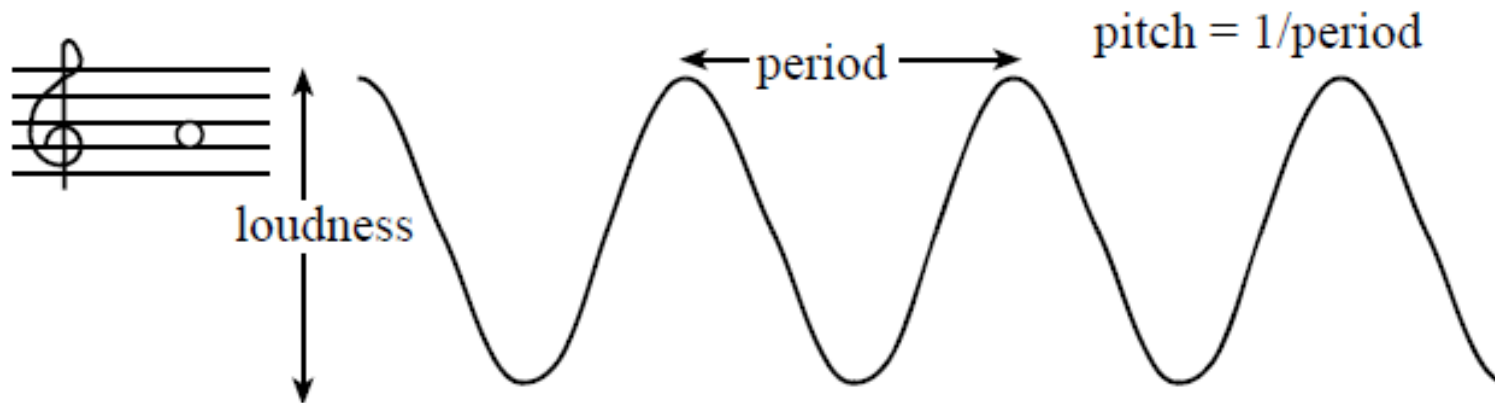
- Sound can be obtained by repeatedly feeding the loudspeaker with a sampled sinusoid.

```
uint16_t SinTable[45] =  
{ 410, 467, 523, 576,  
 627, 673, 714, 749, 778,  
 799, 813, 819, 817, 807,  
 789, 764, 732, 694, 650,  
 602, 550, 495, 438, 381,  
 324, 270, 217, 169, 125,  
 87, 55, 30, 12, 2, 0, 6,  
 20, 41, 70, 105, 146,  
 193, 243, 297, 353};
```



Loudness and pitch

- They are controlled by amplitude and frequency, respectively.



- Humans can hear from about 25 to 20,000 Hz.
- A musical tone (note) is produced by a sinusoid of a particular frequency.

Notes

- Middle A is 440 Hz
 - its sinusoid is completely reproduced in 2.27 ms.
- ♭ indicates a *flat*, i.e., a note lowered by one semitone.

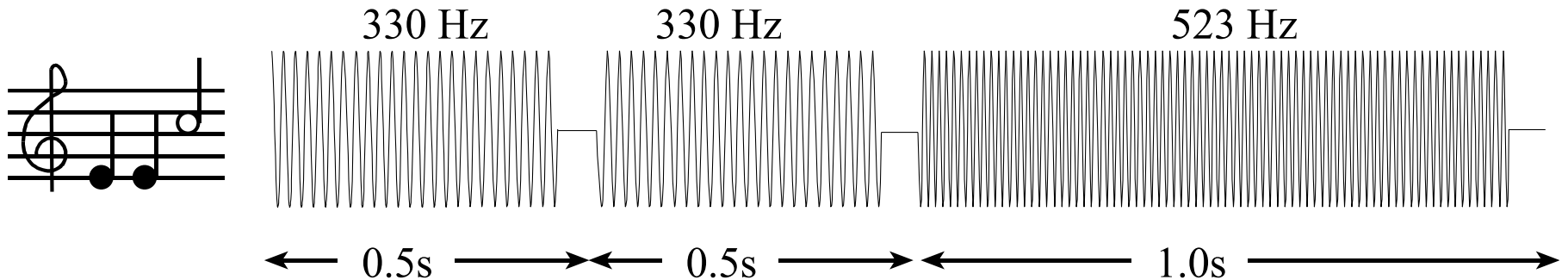
note	solfège	f (Hz)	T (ms)
C	do	262	3.82
D ♭	re ♭	277	3.61
D	re	294	3.40
E ♭	mi ♭	311	3.22
E	mi	330	3.03
F	fa	349	2.87
G ♭	sol ♭	370	2.70
G	sol	392	2.55
A ♭	la ♭	415	2.41
A	<i>la</i>	<i>440</i>	<i>2.27</i>
B ♭	si ♭	466	2.15
B	si	494	2.02
C	do	523	1.91

Synthesizing digital music

- The samples of the sinusoid (corresponding to the tone) are fixed point numbers.
 - the frequencies of notes increase geometrically with a factor of $2^{1/12} \approx 1.059$ for each semitone.
 - e.g., $B \flat = 2^{1/12} A = 2^{1/12} 440 \text{ Hz} = 466 \text{ Hz}$
- A resistor network receives appropriate digital values and produces a pseudo-analog signal.
- A programmable timer regulates the music frequency, by determining when the next value has to be put in output.

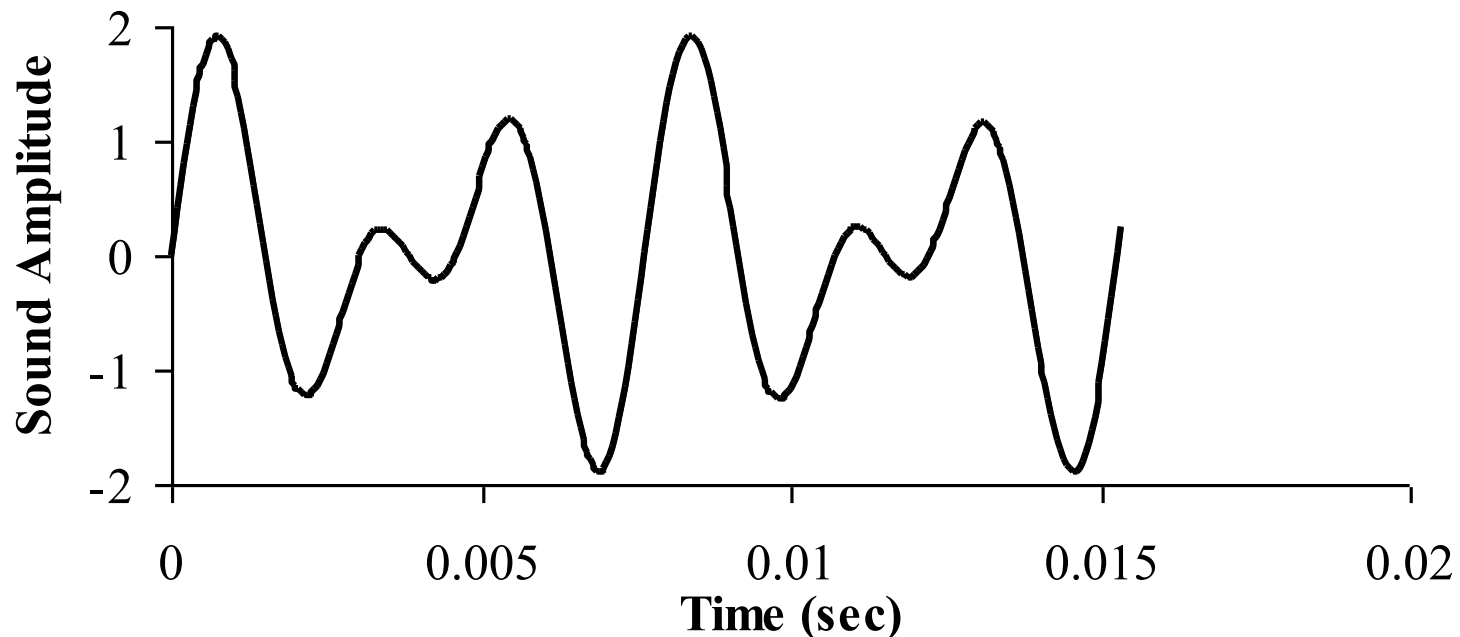
Tempo

- Tempo defines note duration
- Quarter note = 1 beat
- 120 beats/min $\Rightarrow$ $\frac{1}{2}$ s duration



Chord

- It is a group of notes played together (at the same time) for their harmonic consonance
 - superimposed waveforms
 - 262 Hz (low C) and a 392 Hz (G)



Timer handler

- A timer is used to reproduce the sinusoid
- The timer frequency determines the note

```
11 uint16_t SinTable[45] =
12 {
13     410, 467, 523, 576, 627, 673, 714, 749, 778,
14     799, 813, 819, 817, 807, 789, 764, 732, 694,
15     650, 602, 550, 495, 438, 381, 324, 270, 217,
16     169, 125, 87, 55, 30, 12, 2, 0, 6,
17     20, 41, 70, 105, 146, 193, 243, 297, 353
18 };
19
20 void TIMER0_IRQHandler (void)
21 {
22     static int ticks = 0;
23     /* DAC management */
24     LPC_DAC->DACR = (SinTable[ticks]) << 6;
25     ticks ++;
26     if (ticks == 45)
27         ticks=0;
28
29     LPC_TIM0->IR = 1;    /* clear interrupt flag */
30     return;
31 }
32
```